Magnitude/Phase Map of the Four Fermion Sectors

Leptons, quarks and neutrinos in the signal-engineering coordinate — and the neutrinos as the all-pass sector

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Abstract

Do the signal-engineering representations (Doc. 288) help to better capture quarks and neutrinos? The honest answer: *as language and diagnosis yes, as derivation no* – and very differently for the two sectors. Each fermion triplet gets a magnitude coordinate (Koide Q, i.e. $r = \sqrt{6Q-2}$) and a phase coordinate. Computed: only the charged leptons sit at the special point $r = \sqrt{2}$ ($Q = 2/3$); the quarks lie elsewhere ($r \approx 1.54$ resp. 1.76, and scale-dependent), and for neutrinos Q cannot even be formed. The gain is a *unifying map*: leptons and quarks are magnitude-dominated (large hierarchy, small mixing), neutrinos phase-dominated (flat/degenerate magnitude, large mixing – PMNS/CKM weight 0.55 vs 0.035, factor 16). FFGFT's own neutrino hypothesis (equal masses + oscillation from a geometric phase) is then literally the *pure all-pass sector*: magnitude flat, all physics in the phase – and it becomes falsifiable, because the geometric phase must reproduce the measured Δm^2 phases.

.1 The question

Doc. 288 showed that the Z_3-circulant mass operator gives concrete computational levers via the DFT. The natural question: do the same representations reach the hard sectors – quarks and neutrinos? An honest stock-taking is needed first, because the corpus does not treat these sectors like the clean charged leptons.

.2 What the corpus already does (per sector)

- **Charged leptons** (Doc. 006/046/258): the clean case – Foot-Koide form, $Q = 2/3$, $r = \sqrt{2}$, Z_3-circulant. Everything from Doc. 288 applies directly.
- **Quarks** (Doc. 006/005): the same r_i/p_i geometry in principle, but the direct geometric method reaches only $\sim 1.2\%$; for quarks it needs the *extended fractal method* with QCD dynamics and *ML calibration on lattice data* (FLAG 2024). Hence not purely geometric – and quark masses *run* (scale/scheme dependent).
- **Neutrinos** (Doc. 007/047): a *different* mechanism – photon analogy, $m_\nu = \frac{\xi_0^2}{2} m_e \approx 4.54$ meV, all three flavours (nearly) equal in mass, oscillation from *geometric phases* rather than mass differences. Flagged throughout the corpus as **speculative, without empirical basis**.

.3 The (Q, r) coordinate: where the sectors sit

Via $Q = \frac{1}{3} + \frac{r^2}{6}$ (so $r = \sqrt{6Q-2}$) every triplet gets a magnitude coordinate. Computed (`koide_q_sektoren.py`, PDG-near values):

This is already the first answer: *only the charged leptons sit at the special point* $r = \sqrt{2}$. The quark r are not recognisably special (and run with the scale); for neutrinos the absolute scale and ordering are missing, so Q cannot be formed – only $\Delta m^2_{21} \approx 7.5 \cdot 10^{-5}$ and $\Delta m^2_{31} \approx 2.5 \cdot 10^{-3}$ eV^2 are measured.

Table 1: Magnitude coordinate of the three formable triplets. Quark values are scale-dependent (MS-bar).

Sector	Koide Q	$r = \sqrt{6Q-2}$	Position
charged leptons (e, μ, τ)	0.6667	$\sqrt{2} = 1.4142$	special
up quarks (u, c, t)	0.849	1.759	not special
down quarks (d, s, b)	0.730	1.543	not special
neutrinos	—	—	no triplet

.4 The magnitude/phase map

The signal-engineering decomposition (magnitude = mass spectrum, phase = mixing/oscillation) places the four sectors on *one* map:

- **Quarks**: strong mass hierarchy (large magnitude), small mixing (CKM near identity) – *magnitude-dominated*.
- **Charged leptons**: magnitude at a special value ($r = \sqrt{2}$), small phase ($\theta = 2/9$) – *magnitude-dominated*.
- **Neutrinos**: flat magnitude (degenerate), large mixing (PMNS far from identity) – *phase-dominated*.

The CKM/PMNS contrast can be quantified (`ckm_pmns_kontrast.py`, off-diagonal weight of the $|U|^2$ matrix): CKM 0.035 vs PMNS 0.554 – a factor ~ 16. The "small vs large mixing" is thus exactly a *magnitude vs phase* contrast.

.5 Neutrinos as the all-pass sector

Here is the strongest point. FFGFT's own hypothesis – equal masses, oscillation from a geometric phase – is, in magnitude/phase language, literally an *all-pass*: $|H| = 1$ (flat magnitude spectrum), nontrivial phase. A flat magnitude spectrum cannot encode any oscillation; all physics sits in the phase (`neutrino_allpass.py`).

This makes the test *concrete and falsifiable*. The standard oscillation phase is $\Delta\phi_{ij} = 1.267\,\Delta m_{ij}^2 L/E$; at T2K-like L/E ($L = 295$ km, $E = 0.6$ GeV) this is $\Delta\phi_{31} \approx 1.557$ rad and $\Delta\phi_{21} \approx 0.047$ rad. FFGFT's geometric phase (from $\tilde{T} \cdot m = 1$, quantum numbers n, ℓ, j) *must* reproduce exactly these L/E-dependent phases. If it does not, the degeneracy hypothesis is refuted.

.6 What it achieves – and what it does not

Important

Achievement (diagnostic). The magnitude/phase decomposition explains the *division of labour* between the sectors: why the leptons are clean (magnitude at a special value, small phase), the quarks are not (magnitude not special, extreme hierarchy), the neutrinos quite different (magnitude flat, all phase). This is genuine organising knowledge – and it gives the speculative neutrino hypothesis a sharp, falsifiable form.
Boundary (P35). This is better language, a unifying map and a sharper test – *not* a new derivation of quark or neutrino masses. The quark r are not special and scale-dependent, and the quark sector in the corpus relies on QCD plus lattice calibration, not pure geometry. The neutrino sector remains speculative (own flag, Doc. 007)

and in tension with the very well-tested Δm^2 interpretation of oscillation. The clean all-pass map is a compelling reformulation, not new evidence.

Balance: one map, no new sector

The signal-engineering representations capture quarks and neutrinos *better* – but in a precise sense: they give a common magnitude/phase coordinate in which only the charged leptons sit at the special point $r = \sqrt{2}$, the quarks magnitude-dominated beside them, and the neutrinos as a pure phase / all-pass sector. The CKM/PMNS contrast becomes magnitude vs phase (factor 16). And FFGFT's own neutrino hypothesis acquires a falsifiable form: the geometric phase must hit the measured Δm^2 oscillation phases. This is diagnosis and test, not a mass derivation – honestly bounded.